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# DESIGN DATA BOOK — SCREWED JOINTS & FASTENERS

*Standard Engineering Design Data, Formulas & Solved Protocols (IS: 4218 - 1976)*

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## Fastener Joint Classification & Design Index

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| Tap Bolt / Known Diameter | Calculate safe load $P$ or tightening stress $\sigma_t$ | Section 1 & 2     |
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| Offset I-Section Bracket  | Size bolts & I-section arm ( $b = 3t$ )                 | Section 16        |

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|                              |                                                      |            |
|------------------------------|------------------------------------------------------|------------|
| Pillar Crane Base (8-Bolt)   | Size PCD bolts under overturning moment              | Section 17 |
| Flanged Bearing Base         | Size 4 PCD bolts for bearing foundation              | Section 18 |
| Pillar Crane 2-Line Analysis | Determine max load distance $e$ & stress along Y-Y   | Section 19 |
| Solid Forged Bracket         | Size arm dia $D$ (bending+torsion), $d$ & bolt shear | Section 20 |

# SECTION 1: Standard Tensile Capacity Analysis (M 30 Fasteners)

| System Parameters                                                  | Design Protocol                                         |
|--------------------------------------------------------------------|---------------------------------------------------------|
| Nominal Bolt Diameter: $d = 30 \text{ mm}$ (M 30 Coarse Series)    | Retrieve core stress area $A_c$ from Table 11.1         |
| Safe Tensile Stress: $\sigma_t = 42 \text{ MPa} = 42\text{N/mm}^2$ | Calculate safe tensile load capacity $P = A_c \sigma_t$ |
| Preload $F_i = 0$                                                  |                                                         |

## 1. Core Stress Area Lookup ( $A_c$ )

Table 11.1 lookup for M 30 coarse series thread

$$\begin{aligned} A_c &= f(d) \\ &= 561\text{mm}^2 \end{aligned}$$

## 2. Safe Tensile Load Capacity ( $P$ )

Maximum allowable axial tension force without initial tightening

$$\begin{aligned} P &= A_c \times \sigma_t \\ &= 561\text{mm}^2 \times 42\text{N/mm}^2 \\ &= 23562 \text{ N} \\ &= \mathbf{23.562 \text{ kN}} \end{aligned}$$

## 3. Design Output

**Safe Tensile Load  $P = 23.562 \text{ kN}$**



# SECTION 2: Initial Tightening Stress & Preload Analysis (Tap Bolts)

| System Parameters                                       | Design Protocol                                                            |
|---------------------------------------------------------|----------------------------------------------------------------------------|
| Nominal Tap Bolt Size: $d = 24$ mm (M 24 Coarse Series) | Lookup core root diameter $d_c$ from Table 11.1                            |
| Clamped Rigid Machine Joint                             | Calculate initial tightening load $P = 2840d$                              |
| Neglect external separation load                        | Calculate induced tensile stress $\sigma_t = \frac{P}{\frac{\pi}{4}d_c^2}$ |

## 1. Core Root Diameter Lookup ( $d_c$ )

Table 11.1 (coarse series) lookup for M 24 thread

$$d_c = 20.32 \text{ mm}$$

## 2. Initial Tightening Load ( $P$ )

Empirical initial tightening tension equation

$$\begin{aligned} P &= 2840 \cdot d \\ &= 2840 \times 24 \\ &= 68160 \text{ N} \end{aligned}$$

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### 3. Induced Tightening Stress ( $\sigma_t$ )

Initial tension force balance equation

$$\begin{aligned} 68160 &= \frac{\pi}{4}(d_c)^2 \cdot \sigma_t \\ &= \frac{\pi}{4}(20.32)^2 \cdot \sigma_t \\ &= 324 \cdot \sigma_t \\ \sigma_t &= \frac{68160}{324} \\ &= 210\text{N/mm}^2 \\ &= \mathbf{210\text{ MPa}} \end{aligned}$$

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### 4. Design Output

**Tightening Stress  $\sigma_t = 210\text{ MPa}$**

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# SECTION 3: Lifting Eye Bolt Sizing & Thread Selection

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| System Parameters                                                           | Design Protocol                                                          |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Lifting Load: $P = 60 \text{ kN} = 60000 \text{ N}$                         | Equate lifting load to tensile capacity $P = \frac{\pi}{4}d_c^2\sigma_t$ |
| Permissible Tensile Stress: $\sigma_t = 100 \text{ MPa} = 100\text{N/mm}^2$ | Solve required core root diameter $d_c$                                  |
| Thread Series: ISO Metric Coarse                                            | Select standard bolt from Table 11.1                                     |

## 1. Required Core Root Diameter ( $d_c$ )

Axial tension load equation for lifting eye bolt

$$\begin{aligned}P &= \frac{\pi}{4}(d_c)^2 \cdot \sigma_t \\60000 &= \frac{\pi}{4}(d_c)^2 \times 100 \\&= 78.55(d_c)^2 \\(d_c)^2 &= \frac{60000}{78.55} \\&= 763.84 \\d_c &= \sqrt{763.84} \\&= 27.64 \text{ mm}\end{aligned}$$

---

## 2. Standard Thread Selection (Table 11.1)

Select standard coarse series thread with  $d_c \geq 27.64$  mm

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 28.706 \text{ mm}$$

$$d = 33 \text{ mm}$$

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## 3. Design Output

**Select Fastener Designation: M 33 Bolt (d = 33 mm, d<sub>c</sub> = 28.706 mm)**

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## SECTION 3: Lifting Eye Bolt — MECHANICAL DIAGRAM

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### 11.13 Stress due to Combined Forces

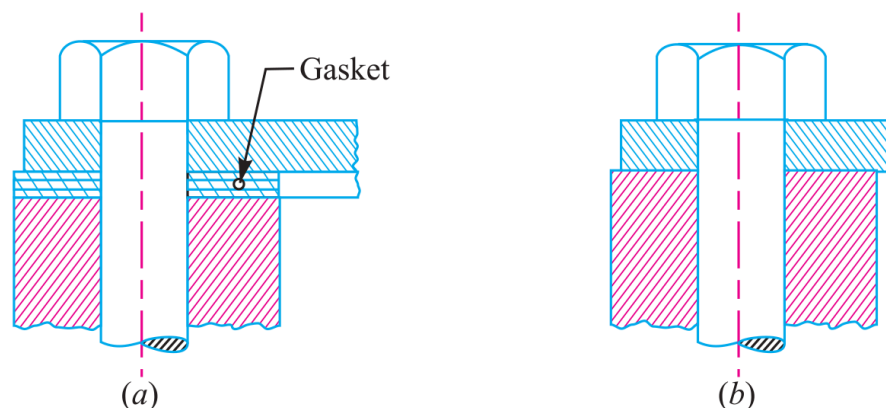


Fig. 11.23

Figure 11.23: Structural Lifting Eye Bolt Configuration and Load Path

# SECTION 4: Flanged Shaft Coupling Fastener Sizing under Torsional Shear

| System Parameters                                                                       | Design Protocol                                                               |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Transmitted Torque: $T = 25 \text{ N} \cdot \text{m} = 25000 \text{ N} \cdot \text{mm}$ | Calculate total tangential shearing load $P_s = \frac{T}{R_p}$                |
| Pitch Circle Radius: $R_p = 30 \text{ mm}$                                              | Equate $P_s$ to shear capacity of $n$ bolts $P_s = n(\frac{\pi}{4}d_c^2)\tau$ |
| Bolt Count: $n = 4$   Allowable Shear Stress: $\tau = 30 \text{ MPa} = 30\text{N/mm}^2$ | Select standard bolt size from Table 11.1                                     |

## 1. Total Shearing Load ( $P_s$ )

Total tangential shearing force acting at PCD

$$\begin{aligned} P_s &= \frac{T}{R_p} \\ &= \frac{25000}{30} \\ &= 833.3 \text{ N} \end{aligned}$$

---

## 2. Required Core Diameter ( $d_c$ )

Solve root diameter for  $n = 4$  bolts carrying shear

$$P_s = n \cdot \left[ \frac{\pi}{4} (d_c)^2 \right] \cdot \tau$$

$$833.3 = 4 \cdot \left[ \frac{\pi}{4} (d_c)^2 \right] \times 30$$

$$= 94.26 (d_c)^2$$

$$(d_c)^2 = \frac{833.3}{94.26}$$

$$= 8.84$$

$$d_c = \sqrt{8.84}$$

$$= 2.97 \text{ mm}$$

---

## 3. Standard Thread Selection (Table 11.1)

Select standard coarse series thread with  $d_c \geq$   
2.97 mm

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 3.141 \text{ mm}$$

$$d = 4 \text{ mm}$$

---

## 4. Design Output

**Select Fastener Designation: M 4 Bolt ( $d = 4 \text{ mm}$ ,  $d_c = 3.141 \text{ mm}$ )**



# SECTION 5: Safety Valve Fulcrum Pin Sizing & Lever Equilibrium

| System Parameters                                                                               | Design Protocol                                        |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Valve Diameter: $D = 100 \text{ mm}$                                                            | Calculate upward steam force $F = \frac{\pi}{4} D^2 p$ |
| Blow-off Steam Pressure: $p = 1.6 \text{ N/mm}^2$                                               | Apply lever moment equilibrium for end load $W$ and    |
| Leverage Ratio: $\frac{L_1}{L_2} = 8$   Permissible Tensile Stress: $\sigma_t = 50 \text{ MPa}$ | fulcrum load $P = F - W$                               |
|                                                                                                 | Size fine metric thread $d_c$ for fulcrum pin          |

## 1. Total Upward Force on Valve ( $F$ )

Total steam pressure load on valve disc

$$\begin{aligned} F &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (100)^2 \times 1.6 \\ &= 12568 \text{ N} \end{aligned}$$

---

## 2. Lever End Load ( $W$ ) & Fulcrum Load ( $P$ )

Static moment equilibrium for Type 1 lever

$$W = \frac{F}{\frac{L_1}{L_2}}$$

$$= \frac{12568}{8}$$

$$= 1571 \text{ N}$$

$$P = F - W$$

$$= 12568 - 1571$$

$$= 10997 \text{ N}$$

---

### 3. Required Core Diameter ( $d_c$ )

Solve root core diameter for fulcrum screw

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$10997 = \frac{\pi}{4}(d_c)^2 \times 50$$

$$= 39.27(d_c)^2$$

$$(d_c)^2 = \frac{10997}{39.27}$$

$$= 280$$

$$d_c = \sqrt{280}$$

$$= 16.7 \text{ mm}$$

---

### 4. Standard Fine Thread Selection (Table 11.1)

Select fine series metric thread with  $d_c \geq$

16.7 mm

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 18.376 \text{ mm}$$

Fine thread: M 20 x 1.5

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### 5. Design Output

**Select Fine Series Thread: M 20 x 1.5 (d = 20 mm, d\_c = 18.376 mm)**





# SECTION 6: Cylinder Head Cover Stud Layout & Leak-Proof Pitch Verification

| System Parameters                                                        | Design Protocol                                                          |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Cylinder Effective Diameter: $D = 350 \text{ mm}$                        | Calculate total upward steam load $P = \frac{\pi}{4} D^2 p$              |
| Maximum Steam Pressure: $p = 1.25 \text{ N/mm}^2$                        | Assume M 24 studs ( $d_c = 20.32 \text{ mm}$ ) to find count             |
| Permissible Stud Stress: $\sigma_t = 33 \text{ MPa} = 33 \text{ N/mm}^2$ | $n = \frac{P}{P_1}$                                                      |
|                                                                          | Verify leak-proof pitch limits $20\sqrt{d_1} \leq p_c \leq 30\sqrt{d_1}$ |

## 1. Total Upward Steam Load ( $P$ )

Total gas pressure load on cylinder cover

$$\begin{aligned} P &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (350)^2 \times 1.25 \\ &= 120265 \text{ N} \end{aligned}$$

---

## 2. Capacity per M 24 Stud ( $P_1$ ) & Stud Count ( $n$ )

Resisting force capacity per M 24 stud ( $d_c = 20.32$  mm)

$$\begin{aligned} P_1 &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \\ &= \frac{\pi}{4} (20.32)^2 \times 33 \\ &= 10700 \text{ N} \end{aligned}$$

$$\begin{aligned} n &= \frac{P}{P_1} \\ &= \frac{120265}{10700} \\ &= 11.24 \Rightarrow n = 12 \text{ studs} \end{aligned}$$

---

## 3. Pitch Circle Diameter ( $D_p$ ) & Circumferential Pitch ( $p_c$ )

Flange wall  $t = 10$  mm, stud hole  $d_1 = 25$  mm

$$\begin{aligned} D_p &= D + 2t + 3d_1 \\ &= 350 + 2(10) + 3(25) \\ &= 445 \text{ mm} \end{aligned}$$

$$\begin{aligned}
 p_c &= \frac{\pi D_p}{n} \\
 &= \frac{\pi \times 445}{12} \\
 &= 116.5 \text{ mm}
 \end{aligned}$$

#### 4. Leak-Proof Pitch Verification

Empirical leak-tight pitch limits ( $d_1$  in mm)

$$\begin{aligned}
 p_{c, \min} &= 20\sqrt{d_1} \\
 &= 20\sqrt{25} \\
 &= 100 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 p_{c, \max} &= 30\sqrt{d_1} \\
 &= 30\sqrt{25} \\
 &= 150 \text{ mm}
 \end{aligned}$$

100 mm  $\leq$  116.5 mm  $\leq$  150 mm (**SAFE & SATISFACTORY**)

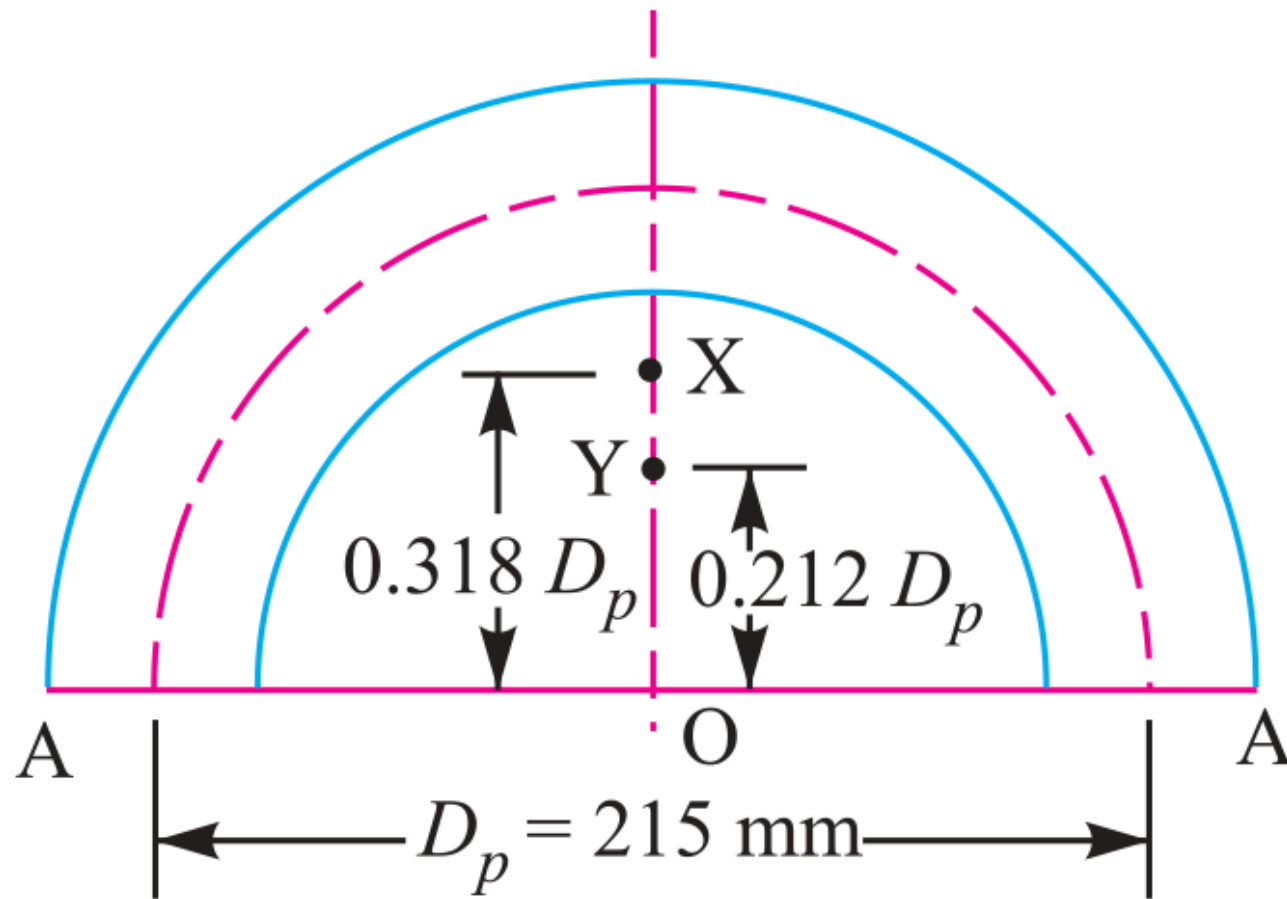
#### 5. Design Output

**Use 12 Studs of M 24 Size (n = 12, M 24 Studs)**

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## **SECTION 6: Cylinder Cover Stud Layout — MECHANICAL DIAGRAM**

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**Fig. 11.27**

**Figure 11.27: Cylinder Head Joint & Stud Pitch Circle Assembly**

# SECTION 7: Pressure Vessel Inspection Cover Plate & Bolt Sizing

| System Parameters                                                                               | Design Protocol                                                    |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Inspection Hole Diameter: $D = 120 \text{ mm}$ ( $r = 60 \text{ mm}$ )                          | Part 1: Calculate shell wall thickness $t$ via Lamé                |
| Internal Vessel Pressure: $p = 6 \text{ N/mm}^2$                                                | Part 2: Determine bolt count $n$ for M 24 and check pitch $p_c$    |
| Plate Tensile Stress: $\sigma_t = 60 \text{ MPa}$   Bolt Stress: $\sigma_{tb} = 40 \text{ MPa}$ | Part 3: Calculate cover plate thickness $t_1$ under bending moment |

## 1. Shell Wall Thickness ( $t$ )

Lamé equation for thick pressure vessels

$$\begin{aligned} t &= r \left[ \sqrt{\frac{\sigma_t + p}{\sigma_t - p}} - 1 \right] \\ &= 60 \left[ \sqrt{\frac{60 + 6}{60 - 6}} - 1 \right] \\ &= 60 \left[ \sqrt{1.222} - 1 \right] \\ &= 6.3 \text{ mm} \end{aligned}$$

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Adopt Shell Thickness  $t = 10 \text{ mm}$

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## 2. Fixing Bolts Sizing & Count ( $n$ )

Upward pressure load  $P$  and M 24 bolt capacity

( $d_c = 20.32 \text{ mm}$ )

$$\begin{aligned} P &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (120)^2 \times 6 \\ &= 67860 \text{ N} \end{aligned}$$

$$\begin{aligned} P_1 &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_{tb} \\ &= \frac{\pi}{4} (20.32)^2 \times 40 \\ &= 12973 \text{ N} \end{aligned}$$

$$\begin{aligned} n &= \frac{P}{P_1} \\ &= \frac{67860}{12973} \\ &= 5.23 \Rightarrow n = 6 \text{ bolts} \end{aligned}$$

---

### 3. PCD & Leak-Proof Pitch Verification

Bolt hole  $d_1 = 25$  mm for M 24 bolts

$$\begin{aligned} D_p &= D + 2t + 3d_1 \\ &= 120 + 2(10) + 3(25) \\ &= 215 \text{ mm} \end{aligned}$$

$$\begin{aligned} p_c &= \frac{\pi D_p}{n} \\ &= \frac{\pi \times 215}{6} \\ &= 112.6 \text{ mm} \end{aligned}$$

100 mm  $\leq$  112.6 mm  $\leq$  150 mm (**SAFE & SATISFACTORY**)

---

### 4. Cover Plate Thickness ( $t_1$ )

Bending moment  $M = 0.053PD_p$  and section modulus  $Z = \frac{1}{6}wt_1^2$

$$\begin{aligned} M &= 0.053PD_p \\ &= 0.053 \times 67860 \times 215 \\ &= 773265 \text{ N} \cdot \text{mm} \end{aligned}$$

$$\begin{aligned} D_o &= D_p + 3d_1 \\ &= 290 \text{ mm} \end{aligned}$$



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$$w = D_o - 2d_1$$

$$= 240 \text{ mm}$$

$$Z = \frac{1}{6} w (t_1)^2$$

$$= \frac{1}{6} \times 240 \times (t_1)^2$$

$$= 40(t_1)^2$$

$$60 = \frac{773265}{40(t_1)^2}$$

$$(t_1)^2 = 322.2$$

$$t_1 = 17.95 \text{ mm}$$

$$= 18 \text{ mm}$$

---

## 5. Design Output

**Final Design: 6 Bolts of M 24 Size, Cover Plate Thickness**

**t<sub>1</sub> = 18 mm**

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# SECTION 8: Preloaded Flange Fasteners with Gasket Compression (Soft Copper)

## System Parameters

Steam Pressure:  $p = 0.7\text{N/mm}^2$   
Number of Bolts:  $n = 12$   
Effective Cylinder Diameter:  $D = 300\text{ mm}$   
Allowable Stress:  $\sigma_t = 100\text{ MPa}$  | Soft Copper Gasket  
Ratio:  $K = 0.5$

## Design Protocol

Calculate external load per bolt  $P_2 = \frac{P_{\text{total}}}{12}$   
Formulate resultant axial load  $P = 2840d + KP_2$   
Solve governing quadratic for nominal diameter  $d$

### 1. External Pressure Load per Bolt ( $P_2$ )

Upward steam pressure load shared by 12 bolts

$$\begin{aligned} P_{\text{total}} &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (300)^2 \times 0.7 \\ &= 49480\text{ N} \end{aligned}$$

$$\begin{aligned}
 P_2 &= \frac{P_{\text{total}}}{n} \\
 &= \frac{49480}{12} \\
 &= 4124 \text{ N}
 \end{aligned}$$

## 2. Resultant Axial Load per Bolt ( $P$ )

Initial tension  $P_1 = 2840d$  plus gasket compression load

$$\begin{aligned}
 P &= P_1 + K \cdot P_2 \\
 &= 2840d + 0.5 \times 4124 \\
 &= 2840d + 2062 \text{ N}
 \end{aligned}$$

## 3. Core Resisting Capacity & Quadratic Equation

Core diameter relation  $d_c = 0.84d$

$$\begin{aligned}
 P &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \\
 &= \frac{\pi}{4} (0.84d)^2 \times 100 \\
 &= 55.4d^2
 \end{aligned}$$

$$55.4d^2 = 2840d + 2062$$

$$55.4d^2 - 2840d - 2062 = 0$$

$$d^2 - 51.3d - 37.2 = 0$$

---

#### 4. Quadratic Diameter Solution ( $d$ )

Quadratic formula evaluation for nominal bolt diameter

$$\begin{aligned}d &= \frac{51.3 \pm \sqrt{(51.3)^2 - 4(1)(-37.2)}}{2} \\&= \frac{51.3 + \sqrt{2631.69 + 148.8}}{2} \\&= \frac{51.3 + 52.73}{2} \\&= \mathbf{52.01 \text{ mm}}\end{aligned}$$

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#### 5. Design Output

**Select Fastener Designation: M 52 Bolt ( $d = 52 \text{ mm}$ )**

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# SECTION 9: Soderberg Fatigue & Preload Design Criterion

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| System Parameters                                                                                         | Design Protocol                                                    |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| $D = 300 \text{ mm}, p = 1.5\text{N/mm}^2, n = 8, \sigma_y = 330 \text{ MPa}, \sigma_e = 240 \text{ MPa}$ | Calculate maximum load $P_1 + K P_2$ & minimum load $P_1$ per bolt |
| Initial Preload: $P_1 = 1.5 P_2$ , Gasket Factor: $K = 0.5$ , “F.S.” = 2                                  | Determine mean load $P_m$ and variable load $P_v$ per bolt         |
|                                                                                                           | Apply Soderberg fatigue equation to solve core diameter $d_c$      |

## 1. Total Steam Load ( $P_2$ ) & Initial Preload ( $P_1$ )

Steam pressure force and initial preload tension

$$\begin{aligned} P_2 &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (300)^2 \times 1.5 \\ &= 106040 \text{ N} \end{aligned}$$

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$$\begin{aligned} P_1 &= 1.5P_2 \\ &= 1.5 \times 106040 \\ &= 159060 \text{ N} \end{aligned}$$

---

## 2. Maximum & Minimum Load per Bolt

Load allocation shared across  $n = 8$  bolts

$$\begin{aligned} P_{\max} &= P_1 + K P_2 \\ &= 159060 + 0.5(106040) \\ &= 212080 \text{ N} \\ P_{\max, \text{ bolt}} &= \frac{212080}{8} \\ &= 26510 \text{ N} \\ P_{\min, \text{ bolt}} &= \frac{159060}{8} \\ &= 19882 \text{ N} \end{aligned}$$

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### 3. Mean ( $P_m$ ) & Variable ( $P_v$ ) Loads per Bolt

Mean and alternating cyclic load components

$$\begin{aligned} P_m &= \frac{P_{\text{max, bolt}} + P_{\text{min, bolt}}}{2} \\ &= \frac{26510 + 19882}{2} \\ &= 23196 \text{ N} \end{aligned}$$

$$\begin{aligned} P_v &= \frac{P_{\text{max, bolt}} - P_{\text{min, bolt}}}{2} \\ &= \frac{26510 - 19882}{2} \\ &= 3314 \text{ N} \end{aligned}$$

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### 4. Mean ( $\sigma_m$ ) & Variable ( $\sigma_v$ ) Stresses

Stress area  $A_c = \frac{\pi}{4}(d_c)^2 = 0.7854(d_c)^2$

$$\begin{aligned} \sigma_m &= \frac{P_m}{A_c} \\ &= \frac{23196}{0.7854(d_c)^2} \\ &= \frac{29534}{(d_c)^2} \text{ N/mm}^2 \end{aligned}$$

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$$\begin{aligned}\sigma_v &= \frac{P_v}{A_c} \\ &= \frac{3314}{0.7854(d_c)^2} \\ &= \frac{4220}{(d_c)^2} \text{N/mm}^2\end{aligned}$$

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## 5. Soderberg Fatigue Failure Equation

Soderberg line equation substitution

$$\sigma_v = \left( \frac{\sigma_e}{\text{F.S.}} \right) \left[ 1 - \frac{\sigma_m \cdot \text{F.S.}}{\sigma_y} \right]$$

$$\frac{4220}{(d_c)^2} = \left( \frac{240}{2} \right) \left[ 1 - \left( \frac{29534}{(d_c)^2} \right) \cdot \left( \frac{2}{330} \right) \right]$$

$$= 120 \left[ 1 - \frac{179.0}{(d_c)^2} \right]$$

$$\frac{4220 + 21480}{(d_c)^2} = 120$$

$$\frac{25700}{(d_c)^2} = 120$$

$$d_c = \sqrt{214.17}$$

$$= 14.63 \text{ mm}$$

---

## 6. Standard Thread Selection (Table 11.1)

Select standard coarse thread with  $d_c \geq 14.63 \text{ mm}$

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 14.933 \text{ mm}$$

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$$d = 18 \text{ mm}$$

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## 7. Design Output

**Select Fastener Designation: M 18 Bolt (d = 18 mm, d<sub>c</sub> = 14.933 mm)**

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# SECTION 10: Boiler Longitudinal Bar Stay Sizing

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| System Parameters                                                         | Design Protocol                                            |
|---------------------------------------------------------------------------|------------------------------------------------------------|
| Pitch of Stays: 350 mm × 350 mm                                           | Calculate area supported by each stay $A = p_s \times p_s$ |
| Steam Pressure: $p = 0.84\text{N/mm}^2$                                   | Calculate steam force $P = A \cdot p$                      |
| Permissible Tensile Stress: $\sigma_t = 56 \text{ MPa} = 56\text{N/mm}^2$ | Solve required core diameter $d_c$ and select bolt size    |

## 1. Area Supported by Each Stay ( $A$ )

Area of boiler plate supported per stay

$$\begin{aligned} A &= p_s \times p_s \\ &= 350 \times 350 \\ &= 122500\text{mm}^2 \end{aligned}$$

## 2. Steam Force Acting on Stay ( $P$ )

Total steam pressure load on supported area

$$\begin{aligned} P &= A \cdot p \\ &= 122500\text{mm}^2 \times 0.84\text{N/mm}^2 \\ &= 102900 \text{ N} \end{aligned}$$

---

### 3. Required Core Root Diameter ( $d_c$ )

Core diameter required for permissible tension stress

$$\begin{aligned} P &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \\ 102900 &= \frac{\pi}{4} (d_c)^2 \times 56 \\ &= 43.98 (d_c)^2 \\ (d_c)^2 &= \frac{102900}{43.98} \\ &= 2339.6 \\ d_c &= \sqrt{2339.6} \\ &= 48.37 \text{ mm} \end{aligned}$$

---

### 4. Standard Thread Selection (Table 11.1)

Select standard coarse thread with  $d_c \geq 48.37$  mm

$$\begin{aligned} d_{c, \text{std}} &\geq d_c \\ d_c &= 49.177 \text{ mm} \\ d &= 56 \text{ mm} \end{aligned}$$

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### 5. Design Output

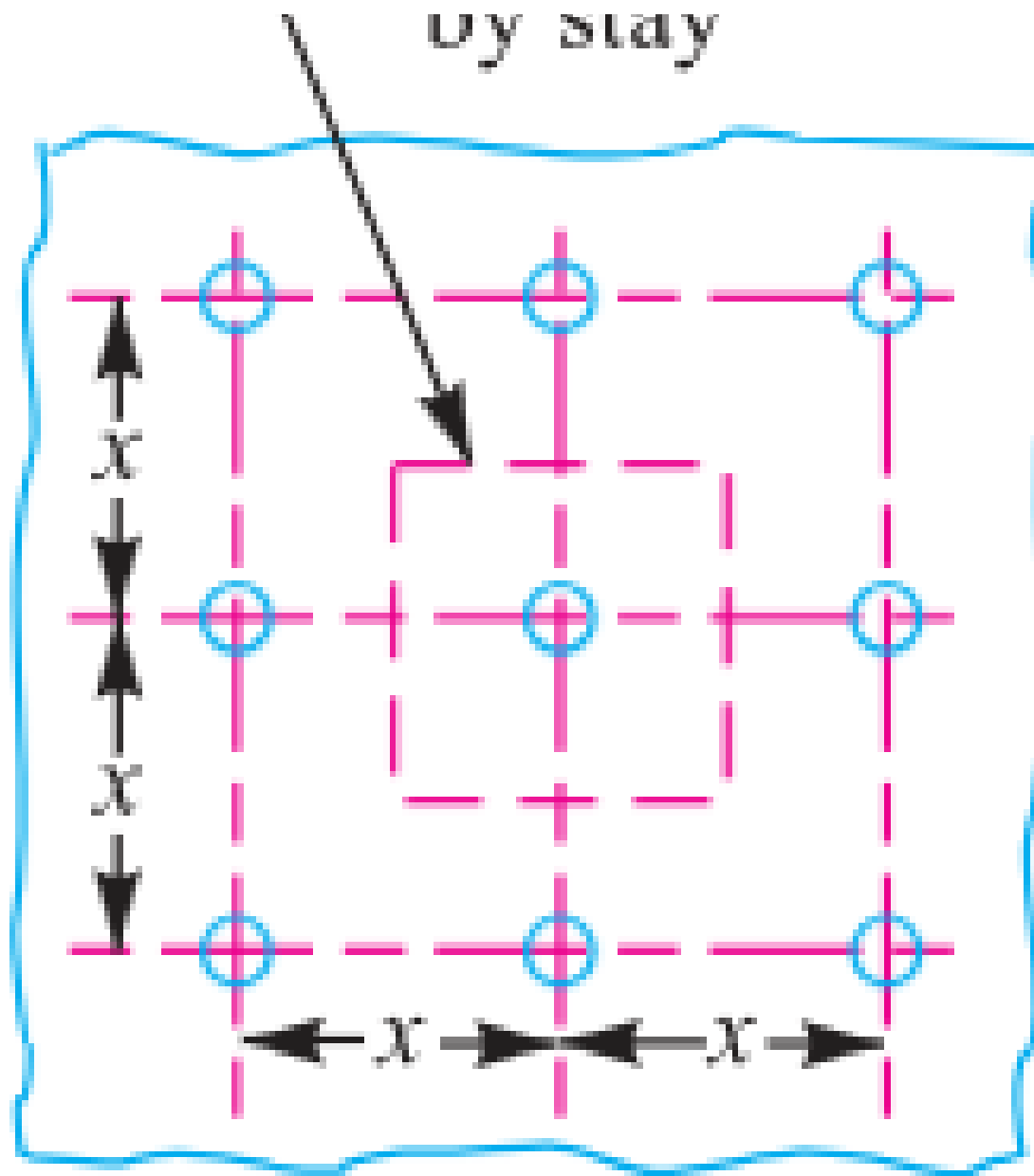
**Select Fastener Designation: M 56 Bolt (d = 56 mm, d\_c = 49.177 mm)**



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## **SECTION 10: Boiler Bar Stay Arrangement — MECHANICAL DIAGRAM**

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# SECTION 11: Turned Shank & Drilled Hole Bolts of Uniform Strength

| System Parameters                                                         | Design Protocol                                                                            |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Nominal Bolt Size: $D_o = 48$ mm (M 48 Coarse Series)                     | Lookup core root diameter $D_c$ from Table 11.1                                            |
| Objective: Determine drilled axial hole diameter $D$ for uniform strength | Equate unthreaded drilled shank area to core root area<br>Solve $D = \sqrt{D_o^2 - D_c^2}$ |

## 1. Core Diameter Lookup ( $D_c$ )

Table 11.1 lookup for M 48 coarse thread

$$\begin{aligned} D_c &= f(D_o) \\ &= 41.795 \text{ mm} \end{aligned}$$

## 2. Area Equality Condition for Uniform Strength

Equate unthreaded drilled shank area to root area

$$\begin{aligned} \frac{\pi}{4}(D_o^2 - D^2) &= \frac{\pi}{4}D_c^2 \\ D_o^2 - D^2 &= D_c^2 \end{aligned}$$



---

### 3. Drilled Axial Hole Diameter ( $D$ )

Solve central axial hole diameter

$$\begin{aligned} D &= \sqrt{D_o^2 - D_c^2} \\ &= \sqrt{(48)^2 - (41.795)^2} \\ &= \sqrt{2304 - 1746.82} \\ &= \sqrt{557.18} \\ &= \mathbf{23.64 \text{ mm}} \end{aligned}$$

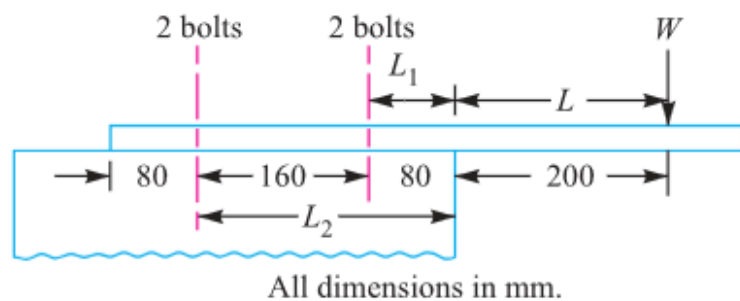
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### 4. Design Output

**Drill Central Axial Hole of Diameter  $D = 23.64 \text{ mm}$**

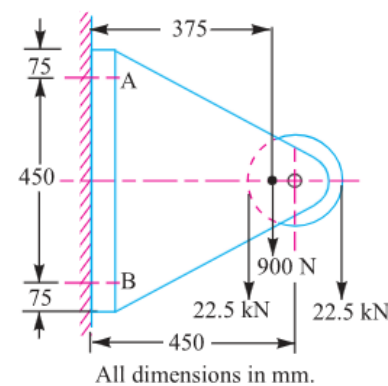
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## SECTION 11: Uniform Strength Fasteners — MECHANICAL DIAGRAM

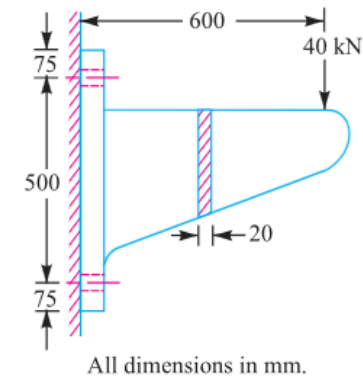


**Fig. 11.48**

**Figure 11.48: Turned Shank Uniform Strength Bolt**



**Fig. 11.49**



**Fig. 11.50**

**Figure 11.49 & 11.50: Core-Drilled Hole Uniform Strength Bolt**

# SECTION 12: Wall Brackets under Parallel Eccentric Tensile Loading

| System Parameters                                                               | Design Protocol                                                 |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Load: $W = 30 \text{ kN} = 30000 \text{ N}$                                     | Calculate direct tensile load $W_{t1} = \frac{W}{n}$            |
| Eccentricity Arm: $L = 500 \text{ mm}$                                          | Calculate unit secondary load $w = \frac{WL}{2(L_1^2 + L_2^2)}$ |
| Distances: $L_1 = 80 \text{ mm}$ , $L_2 = 250 \text{ mm}$                       | Calculate total tension $W_t = W_{t1} + wL_2$ and core          |
| Bolt Count: $n = 4$ (2 per row)   Allowable Stress: $\sigma_t = 60 \text{ MPa}$ | size $d_c$                                                      |

## 1. Direct Tensile Load per Bolt ( $W_{t1}$ )

Direct load shared equally by 4 bolts

$$\begin{aligned} W_{t1} &= \frac{W}{n} \\ &= \frac{30000}{4} \\ &= 7500 \text{ N} \\ &= 7.5 \text{ kN} \end{aligned}$$

---

## 2. Secondary Tensile Load on Upper Bolts

( $W_{t2}$ )

Moment equilibrium load factor  $w$

$$\begin{aligned}w &= \frac{W \cdot L}{2(L_1^2 + L_2^2)} \\&= \frac{30 \times 500}{2(80^2 + 250^2)} \\&= \frac{15000}{2(6400 + 62500)} \\&= 0.10886 \text{ kN/mm}\end{aligned}$$

$$\begin{aligned}W_{t2} &= w \cdot L_2 \\&= 0.10886 \times 250 \\&= 27.215 \text{ kN} \\&= 27215 \text{ N}\end{aligned}$$

---

## 3. Total Maximum Tensile Load ( $W_t$ )

Combined direct and secondary tension on critical top bolts

$$\begin{aligned}W_t &= W_{t1} + W_{t2} \\&= 7.5 + 27.215 \\&= 34.715 \text{ kN} \\&= 34715 \text{ N}\end{aligned}$$

---

#### 4. Required Core Diameter ( $d_c$ )

Solve root diameter for top row fasteners

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$34715 = \frac{\pi}{4}(d_c)^2 \times 60$$

$$= 47.12(d_c)^2$$

$$(d_c)^2 = \frac{34715}{47.12}$$

$$= 736.7$$

$$d_c = \sqrt{736.7}$$

$$= 27.14 \text{ mm}$$

---

#### 5. Standard Thread Selection (Table 11.1)

Select standard coarse thread with  $d_c \geq 27.14 \text{ mm}$

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 28.706 \text{ mm}$$

$$d = 33 \text{ mm}$$

---

#### 6. Design Output

**Select Fastener Designation: M 33 Bolt (d = 33 mm, d\_c = 28.706 mm)**



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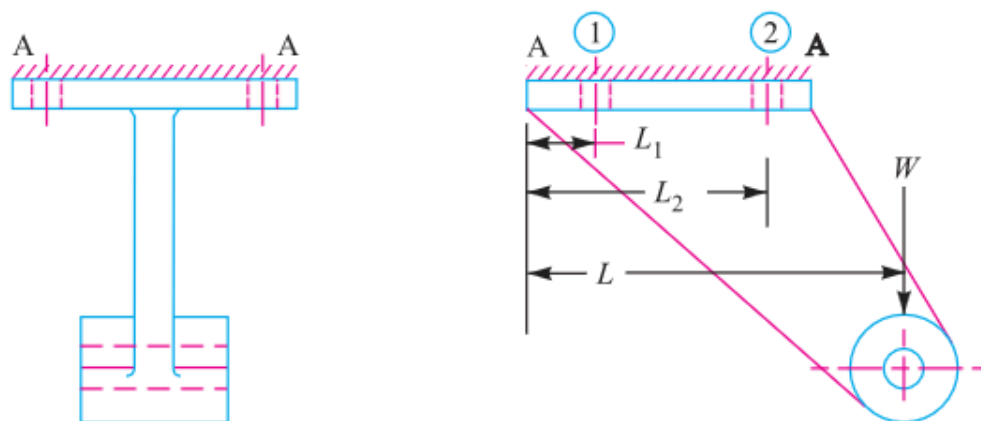
## SECTION 12: Wall Bracket (Parallel Load) — MECHANICAL DIAGRAM

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### 11.19 Eccentric Load Acting Parallel to the Axis of Bolts

Consider a bracket having a rectangular base bolted to a wall by means of four bolts as shown in Fig. 11.31. A little consideration will show that each bolt is subjected to a direct tensile load of

$$W_{t1} = \frac{W}{n}, \text{ where } n \text{ is the number of bolts.}$$



**Fig. 11.31.** Eccentric load acting parallel to the axis of bolts.

**Figure 11.31: Wall Bracket under Parallel Eccentric Load**

# SECTION 13: Crane Runway T-Bracket Stress Analysis & Fastening Bolts

| System Parameters                                                                      | Design Protocol                                                                                               |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Wheel Load: $W = 15 \text{ kN} = 15000 \text{ N}$                                      | Part 1: Calculate centroid $\bar{y}$ , moment of inertia $I_{GG}$ , and max arm stresses $\sigma_t, \sigma_c$ |
| Fastening Bolts: $d = 25 \text{ mm}$ ( $n = 4$ )                                       | Part 2: Calculate direct & secondary bolt tension to find bolt stress $\sigma_{tb}$                           |
| T-Section: Flange $135 \times 25$ , Web $175 \times 25$ , Total Depth $200 \text{ mm}$ |                                                                                                               |

## 1. Centroid ( $\bar{y}$ ) & Section Area ( $A$ )

T-section geometry evaluation

$$A_1 = 135 \times 25 = 3375\text{mm}^2$$

$$A_2 = 175 \times 25 = 4375\text{mm}^2$$

$$\begin{aligned} A &= A_1 + A_2 \\ &= 7750\text{mm}^2 \end{aligned}$$



$$\begin{aligned}
\bar{y} &= \frac{A_1 y_1 + A_2 y_2}{A} \\
&= \frac{3375 \times 12.5 + 4375 \times 112.5}{7750} \\
&= 68.95 \text{ mm} \\
&\approx 69 \text{ mm} \\
y_1 &= 69 \text{ mm} \quad (\text{Top}) \\
y_2 &= 200 - 69 = 131 \text{ mm} \quad (\text{Bottom})
\end{aligned}$$

## 2. Moment of Inertia ( $I_{GG}$ ) & Section Moduli ( $Z_1, Z_2$ )

Parallel axis theorem for T-section

$$\begin{aligned}
I_{GG} &= \sum (I_i + A_i d_i^2) \\
&= \left[ \frac{135(25)^3}{12} + 3375(56.5)^2 \right] + \left[ \frac{25(175)^3}{12} + 4375(43.5)^2 \right] \\
&= 30.4 \times 10^6 \text{ mm}^4
\end{aligned}$$

$$\begin{aligned}
 Z_1 &= \frac{I_{GG}}{y_1} \\
 &= \frac{30.4 \times 10^6}{69} \\
 &= 440.6 \times 10^3 \text{ mm}^3
 \end{aligned}$$

$$\begin{aligned}
 Z_2 &= \frac{I_{GG}}{y_2} \\
 &= \frac{30.4 \times 10^6}{131} \\
 &= 232 \times 10^3 \text{ mm}^3
 \end{aligned}$$

### 3. Bracket Arm Stresses at Section X-X

Bending moment  $M = 15000(200 + 69) = 4.035 \times 10^6 \text{ N} \cdot \text{mm}$

$$\begin{aligned}
 \sigma_{b1} &= \frac{M}{Z_1} \\
 &= \frac{4.035 \times 10^6}{440.6 \times 10^3} \\
 &= 9.16 \text{ N/mm}^2
 \end{aligned}$$

$$\begin{aligned}\sigma_{b2} &= \frac{M}{Z_2} \\ &= \frac{4.035 \times 10^6}{232 \times 10^3} \\ &= 17.4 \text{ N/mm}^2\end{aligned}$$

$$\begin{aligned}\sigma_{t1} &= \frac{W}{A} \\ &= \frac{15000}{7750} \\ &= 1.94 \text{ N/mm}^2\end{aligned}$$

$$\begin{aligned}\sigma_t &= \sigma_{b1} + \sigma_{t1} \\ &= 9.16 + 1.94 \\ &= 11.1 \text{ MPa (Tensile)}\end{aligned}$$

$$\begin{aligned}\sigma_c &= \sigma_{b2} - \sigma_{t1} \\ &= 17.4 - 1.94 \\ &= 15.46 \text{ MPa (Compressive)}\end{aligned}$$

---

#### 4. Fastening Bolt Loads ( $W_{t1}$ , $W_{t2}$ , $W_t$ )

Tilting edge  $\mathbb{E}$ :  $L_1 = 50$  mm,  $L_2 = 375$  mm,  $L = 525$  mm

$$\begin{aligned} W_{t1} &= \frac{W}{n} \\ &= \frac{15000}{4} \\ &= 3750 \text{ N} \end{aligned}$$

$$\begin{aligned} w &= \frac{W \cdot L}{2(L_1^2 + L_2^2)} \\ &= \frac{15000 \times 525}{2(50^2 + 375^2)} \\ &= \frac{7875000}{286250} \\ &= 27.5 \text{ N/mm} \end{aligned}$$

$$\begin{aligned} W_{t2} &= w \cdot L_2 \\ &= 27.5 \times 375 \\ &= 10312.5 \text{ N} \end{aligned}$$

$$\begin{aligned}
 W_t &= W_{t1} + W_{t2} \\
 &= 3750 + 10312.5 \\
 &= 14062.5 \text{ N}
 \end{aligned}$$

## 5. Maximum Tensile Stress in Fastening Bolts ( $\sigma_{tb}$ )

Bolt core diameter  $d_c = 0.84d = 0.84 \times 25 = 21 \text{ mm}$

$$\begin{aligned}
 \sigma_{tb} &= \frac{W_t}{\frac{\pi}{4}(d_c)^2} \\
 &= \frac{14062.5}{\frac{\pi}{4}(21)^2} \\
 &= \frac{14062.5}{346.36} \\
 &= 40.6 \text{ MPa}
 \end{aligned}$$

## 6. Design Output

**Arm Tensile  $\sigma_t = 11.1 \text{ MPa}$ , Arm Compressive  $\sigma_c = 15.46 \text{ MPa}$ , Bolt Stress  $\sigma_{tb} = 40.6 \text{ MPa}$**

## SECTION 13: Crane Runway T-Bracket — MECHANICAL DIAGRAM

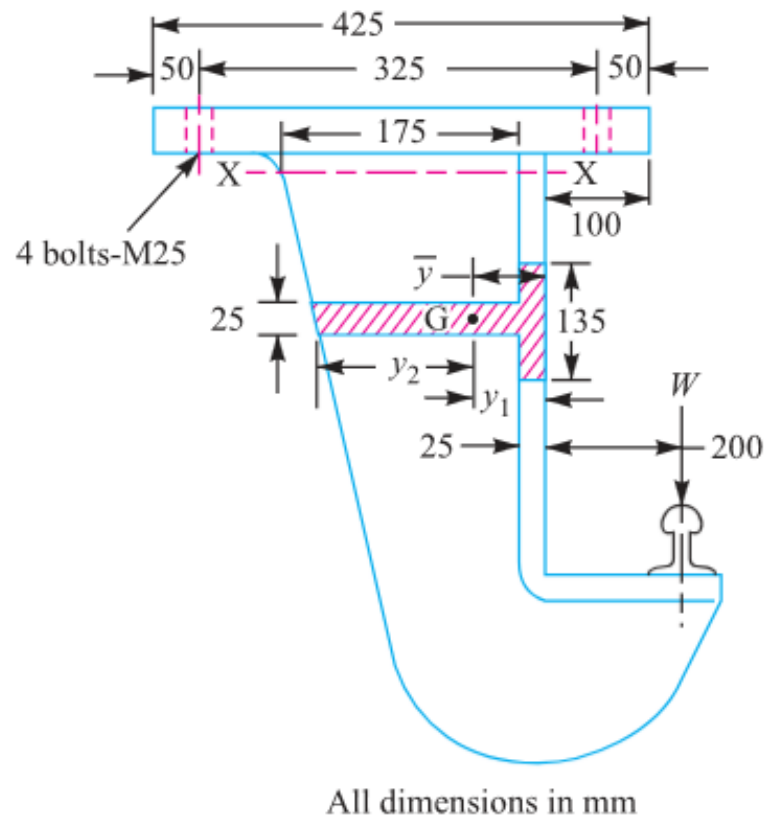


Fig. 11.32

Figure 11.32: Crane Runway T-Bracket Structure

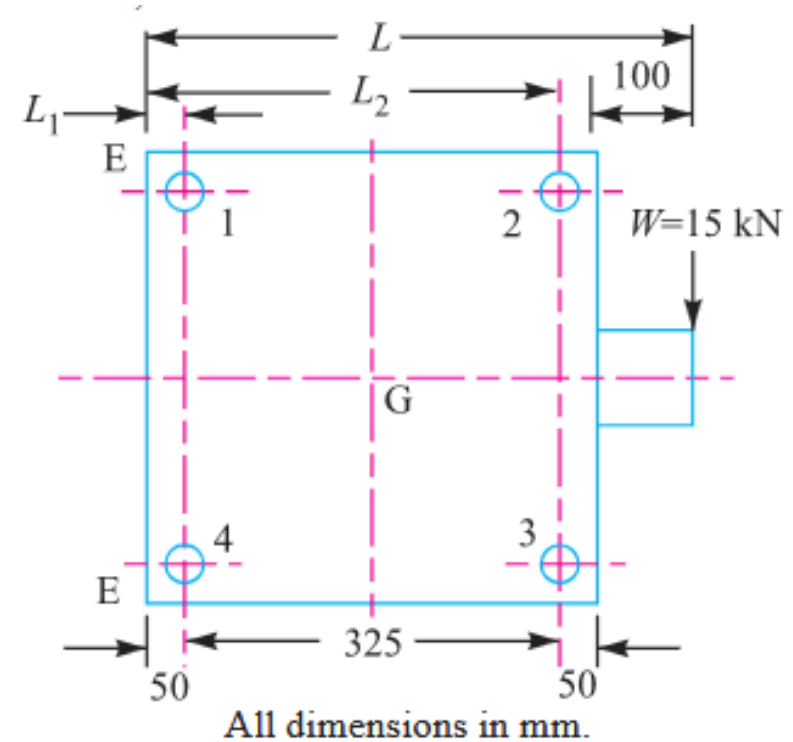


Fig. 11.33

Figure 11.33: T-Section Stress Distribution Diagram

# SECTION 14: Travelling Crane Bracket Sizing (Combined Shear & Tension)

| System Parameters                                                             | Design Protocol                                                                                                                   |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Load: $W = 12 \text{ kN} = 12000 \text{ N}$                                   | Part 1: Calculate direct shear $W_s$ , secondary tension $W_t$ , and equivalent load $W_{te}$ via Maximum Principal Stress Theory |
| Load Arm: $L = 400 \text{ mm}$                                                |                                                                                                                                   |
| Distances: $L_1 = 50 \text{ mm}$ , $L_2 = 375 \text{ mm}$                     |                                                                                                                                   |
| Permissible Tensile Stress: $\sigma_t = 84 \text{ MPa}$   Bolt Count: $n = 4$ | Part 2: Calculate arm thickness $t$ for depth $b = 250 \text{ mm}$                                                                |

## 1. Direct Shear ( $W_s$ ) & Secondary Tension ( $W_t$ )

Primary shear load and secondary tilting tension per bolt

$$\begin{aligned} W_s &= \frac{W}{n} \\ &= \frac{12000}{4} \\ &= 3000 \text{ N} \\ &= 3 \text{ kN} \end{aligned}$$

$$\begin{aligned}
 W_t &= \frac{W \cdot L \cdot L_2}{2(L_1^2 + L_2^2)} \\
 &= \frac{12 \times 400 \times 375}{2(50^2 + 375^2)} \\
 &= \frac{1800000}{286250} \\
 &= 6.288 \text{ kN} \\
 &= 6288 \text{ N}
 \end{aligned}$$

## 2. Equivalent Tensile Load ( $W_{te}$ )

Maximum Principal Stress Theory for combined shear & tension

$$\begin{aligned}
 W_{te} &= \frac{1}{2} \left[ W_t + \sqrt{W_t^2 + 4W_s^2} \right] \\
 &= \frac{1}{2} \left[ 6.29 + \sqrt{(6.29)^2 + 4(3)^2} \right] \\
 &= \frac{1}{2} [6.29 + 8.69] \\
 &= 7.49 \text{ kN} \\
 &= 7490 \text{ N}
 \end{aligned}$$



---

### 3. Required Core Diameter ( $d_c$ ) & Bolt Selection

Table 11.1 lookup for metric coarse thread

$$W_{te} = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$7490 = \frac{\pi}{4}(d_c)^2 \times 84$$

$$= 65.97(d_c)^2$$

$$(d_c)^2 = \frac{7490}{65.97}$$

$$= 113.5$$

$$d_c = \sqrt{113.5}$$

$$= 10.65 \text{ mm}$$

---

### 4. Rectangular Cross-Section of Bracket Arm ( $t \times b$ )

Bending moment  $M = W \cdot L = 12 \times 10^3 \times 400 = 4.8 \times 10^6 \text{ N} \cdot \text{mm}$

$$Z = \frac{1}{6} \cdot t \cdot b^2$$

---

$$\sigma_t = \frac{M}{Z}$$

$$84 = \frac{4.8 \times 10^6}{\frac{1}{6} \cdot t \cdot b^2}$$

$$t \cdot b^2 = \frac{28.8 \times 10^6}{84}$$

$$= 342857 \text{ mm}^3$$

$$t = \frac{342857}{(250)^2}$$

$$= \frac{342857}{62500}$$

$$= 5.5 \text{ mm}$$

---

## 5. Design Output

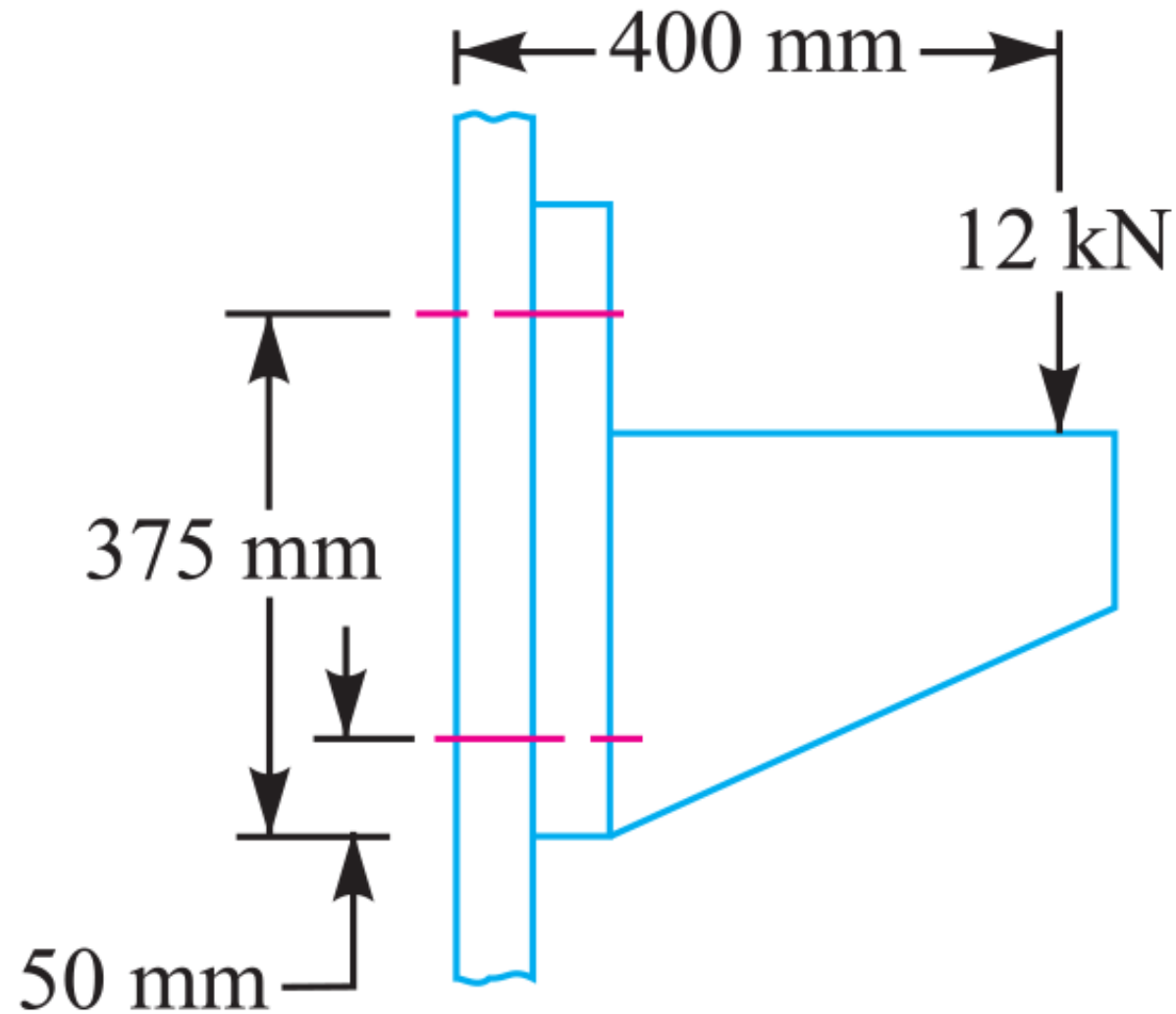
**Final Design: M 14 Bolts, Bracket Arm Dimensions 5.5 mm x 250 mm**

---

---

## **SECTION 14: Travelling Crane Bracket — MECHANICAL DIAGRAM**

---



**Fig. 11.35**

# SECTION 15: Inclined Load Bracket Fastener Sizing & Arm Thickness Analysis

| System Parameters                                                                                 | Design Protocol                                                                           |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Inclined Load: $W = 40 \text{ kN} = 40000 \text{ N}$ at $\theta = 60^{\text{deg}}$ to vertical    | Part 1: Resolve load into $W_H$ , $W_V$ and net moment<br>$T_{\text{net}} = T_V - T_H$    |
| Stresses: $\sigma_t = 70 \text{ MPa}$ , $\tau = 50 \text{ MPa}$ , $\sigma_c = 105 \text{ MPa}$    | Part 2: Calculate bolt tension $W_t$ , equivalent load $W_{te}$ , and size $d_c$          |
| Arm Depth: $b = 130 \text{ mm}$   Bolts: $n = 4$ , $L_1 = 60 \text{ mm}$ , $L_2 = 180 \text{ mm}$ | Part 3: Calculate arm thickness $t$ via principal tensile stress $\sigma_{t(\text{max})}$ |

## 1. Force Resolution & Net Turning Moment ( $T_{\text{net}}$ )

Horizontal and vertical force components

$$\begin{aligned} W_H &= W \sin \theta \\ &= 40 \sin 60^{\text{deg}} \\ &= 34.64 \text{ kN} \\ &= 34640 \text{ N} \end{aligned}$$

$$W_V = W \cos \theta$$

$$= 40 \cos 60^{\text{deg}}$$

$$= 20 \text{ kN}$$

$$= 20000 \text{ N}$$

$$T_H = 34640 \times 20$$

$$= 692.8 \times 10^3 \text{ N} \cdot \text{mm}$$

$$T_V = 20000 \times 175$$

$$= 3500 \times 10^3 \text{ N} \cdot \text{mm}$$

$$T_{\text{net}} = T_V - T_H$$

$$= (3500 - 692.8) \times 10^3$$

$$= 2807.2 \times 10^3 \text{ N} \cdot \text{mm}$$

## 2. Fixing Bolts Sizing ( $W_t$ , $W_s$ , $W_{te}$ , $d_c$ )

Direct & secondary load combination on critical top bolts

$$W_{t1} = \frac{W_H}{n}$$

$$= 8660 \text{ N}$$

$$W_s = \frac{W_V}{n}$$

$$= 5000 \text{ N}$$

$$w = \frac{T_{\text{net}}}{2(L_1^2 + L_2^2)}$$

$$= \frac{2807.2 \times 10^3}{2(60^2 + 180^2)}$$

$$= 38.99 \text{ N/mm}$$

$$\approx 39 \text{ N/mm}$$

$$W_{t2} = w \cdot L_2$$

$$= 39 \times 180$$

$$= 7020 \text{ N}$$

$$W_t = W_{t1} + W_{t2}$$

$$= 8660 + 7020$$

$$= 15680 \text{ N}$$

$$\begin{aligned}
 W_{te} &= \frac{1}{2} \left[ W_t + \sqrt{W_t^2 + 4W_s^2} \right] \\
 &= \frac{1}{2} \left[ 15680 + \sqrt{(15680)^2 + 4(5000)^2} \right] \\
 &= 17140 \text{ N} \\
 (d_c)^2 &= \frac{17140}{55} \\
 &= 311.64 \\
 d_c &= 17.65 \text{ mm}
 \end{aligned}$$

### 3. Bracket Arm Thickness ( $t$ via Principal Stress)

Stresses in upper arm fibre for  $b = 130 \text{ mm}$  ( $Z = 2817t$ )

$$\begin{aligned}
 \sigma_{t1} &= \frac{W_H}{A} \\
 &= \frac{34640}{130t} \\
 &= \frac{266.5}{t}
 \end{aligned}$$



$$\begin{aligned}\sigma_{t2} &= \frac{M_H}{Z} \\ &= \frac{34640 \times 35}{2817t} \\ &= \frac{430.4}{t}\end{aligned}$$

$$\begin{aligned}\sigma_{t3} &= \frac{M_V}{Z} \\ &= \frac{20000 \times 200}{2817t} \\ &= \frac{1420}{t}\end{aligned}$$

$$\begin{aligned}\sigma_t &= \sigma_{t1} + \sigma_{t2} + \sigma_{t3} \\ &= \frac{266.5 + 430.4 + 1420}{t} \\ &= \frac{2116.9}{t}\end{aligned}$$

$$\begin{aligned}\tau &= \frac{W_V}{A} \\ &= \frac{20000}{130t} \\ &= \frac{154}{t}\end{aligned}$$

$$\begin{aligned}\sigma_{t(\max)} &= \frac{\sigma_t}{2} + \frac{1}{2}\sqrt{\sigma_t^2 + 4\tau^2} \\ &= \frac{1058.45}{t} + \frac{1069.6}{t} \\ &= \frac{2128.05}{t} \\ 70 &= \frac{2128.05}{t} \\ t &= 30.4 \text{ mm} \\ &= \mathbf{31 \text{ mm}}\end{aligned}$$

#### 4. Design Output

**Final Design: M 22 Bolts, Bracket Arm Thickness t = 31 mm**



SECTION 15: Inclined Load Bracket — MECHANICAL DIAGRAM

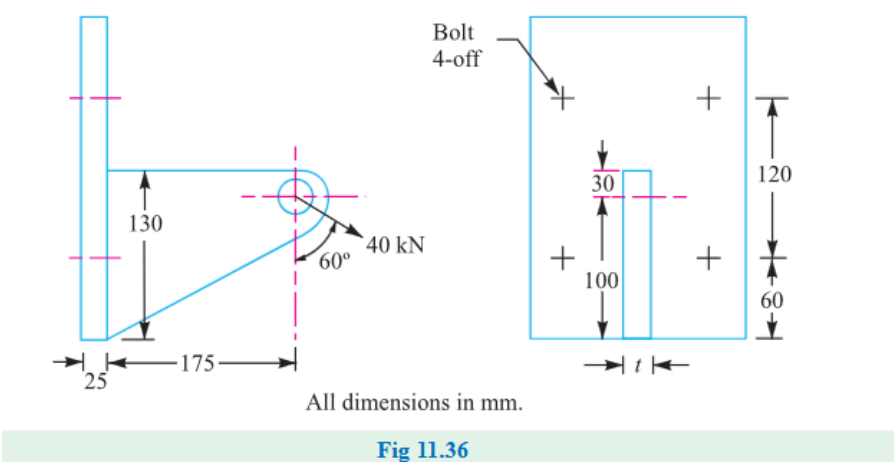


Figure 11.36: Wall Bracket under Inclined Load

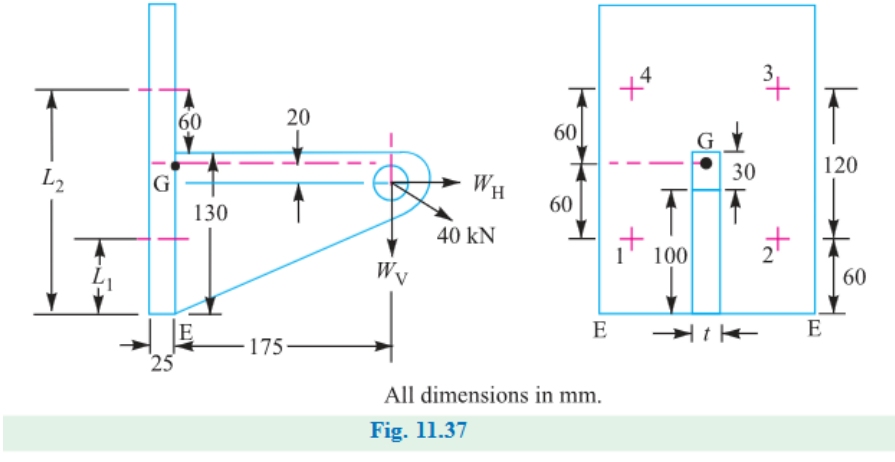


Figure 11.37: Load Resolution & Moment Arm Diagram

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# SECTION 16: Offset I-Section Bracket under Inclined Loading

---

| System Parameters                                                                          | Design Protocol                                                                         |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Pull Load: $W = 10 \text{ kN} = 10000 \text{ N}$ at $\theta = 60^{\text{deg}}$ to vertical | Part 1: Resolve load into $W_H$ , $W_V$ and net moment                                  |
| Stresses: $\sigma_t = 100 \text{ MPa}$ , $\tau = 60 \text{ MPa}$                           | $T_{\text{net}} = T_V - T_H$                                                            |
| Bolt Count: $n = 4$   I-Section Flange Width: $b = 3t$                                     | Part 2: Calculate bolt tension $W_t$ , equivalent load $W_{\text{te}}$ , and size $d_c$ |
|                                                                                            | Part 3: Calculate I-section arm thickness $t$ and width $b = 3t$                        |

## 1. Force Resolution & Net Turning Moment ( $T_{\text{net}}$ )

Component forces and net moment about tilting edge  $E$

$$\begin{aligned} W_H &= W \sin \theta \\ &= 10 \sin 60^{\text{deg}} \\ &= 8.66 \text{ kN} \\ &= 8660 \text{ N} \end{aligned}$$

$$W_V = W \cos \theta$$

$$= 10 \cos 60^{\text{deg}}$$

$$= 5 \text{ kN}$$

$$= 5000 \text{ N}$$

$$T_H = 433 \text{ N} \cdot \text{m} \quad (\text{Clockwise})$$

$$T_V = 1500 \text{ N} \cdot \text{m} \quad (\text{Counterclockwise})$$

$$T_{\text{net}} = T_V - T_H$$

$$= 1500 - 433$$

$$= 1067 \text{ N} \cdot \text{m}$$

## 2. Fixing Bolts Sizing ( $W_t$ , $W_s$ , $W_{te}$ , $d_c$ )

Bolt distances  $L_1 = 0.0375 \text{ m}$ ,  $L_2 = 0.2125 \text{ m}$

$$W_{t1} = \frac{W_H}{n}$$

$$= 2165 \text{ N}$$

$$W_s = \frac{W_V}{n}$$

$$= 1250 \text{ N}$$

---

$$\begin{aligned}w &= \frac{T_{\text{net}}}{2(L_1^2 + L_2^2)} \\&= \frac{1067}{2(0.0375^2 + 0.2125^2)} \\&= 11470 \text{ N/m}\end{aligned}$$

$$\begin{aligned}W_{t2} &= w \cdot L_2 \\&= 11470 \times 0.2125 \\&= 2435 \text{ N}\end{aligned}$$

$$\begin{aligned}W_t &= W_{t1} + W_{t2} \\&= 2165 + 2435 \\&= 4600 \text{ N}\end{aligned}$$

$$\begin{aligned}
 W_{te} &= \frac{1}{2} \left[ W_t + \sqrt{W_t^2 + 4W_s^2} \right] \\
 &= \frac{1}{2} \left[ 4600 + \sqrt{(4600)^2 + 4(1250)^2} \right] \\
 &= 4920 \text{ N} \\
 (d_c)^2 &= \frac{4920}{78.55} \\
 &= 62.63 \\
 d_c &= 7.91 \text{ mm}
 \end{aligned}$$

### 3. Dimensions of I-Section Bracket Arm ( $t$ and $b = 3t$ )

Cross-section area  $A = 9t^2$ , section modulus  $Z = 10.7t^3$

$$\begin{aligned}
 \sigma_{t1} &= \frac{W_H}{A} \\
 &= \frac{8660}{9t^2} \\
 &= \frac{962}{t^2}
 \end{aligned}$$



$$\begin{aligned}\sigma_{t2} &= \frac{M_H}{Z} \\ &= \frac{433 \times 10^3}{10.7t^3} \\ &= \frac{40.5 \times 10^3}{t^3}\end{aligned}$$

$$\begin{aligned}\sigma_{t3} &= \frac{M_V}{Z} \\ &= \frac{1500 \times 10^3}{10.7t^3} \\ &= \frac{140.2 \times 10^3}{t^3}\end{aligned}$$

$$\begin{aligned}\sigma_t &= \sigma_{t1} - \sigma_{t2} + \sigma_{t3} \\ &= \frac{962}{t^2} + \frac{99.7 \times 10^3}{t^3}\end{aligned}$$

$$100 = \frac{962}{t^2} + \frac{99.7 \times 10^3}{t^3} \Rightarrow t = 10.4 \text{ mm}$$

$$b = 3t = 31.2 \text{ mm}$$

---

#### 4. Design Output

**Final Design: M 10 Bolts, I-Section Arm Dimensions  $t = 10.4$  mm,  $b = 31.2$  mm**

---

# SECTION 17: Pillar Crane Circular Base Foundation Flange Sizing

| System Parameters                                                                                                                  | Design Protocol                                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Bolt Count: $n = 8$                                                                                                                | Calculate distance from tilting edge $A - A$ : $L = e - R$              |
| Bolt Circle Diameter: $d = 1.6 \text{ m} \Rightarrow r = 0.8 \text{ m}$                                                            |                                                                         |
| Pillar Base Diameter: $D = 2 \text{ m} \Rightarrow R = 1 \text{ m}$                                                                | Calculate maximum tensile load $W_t$ on critical bolt using PCD formula |
| Crane Load: $W = 100 \text{ kN} = 100000 \text{ N}$ at distance $e = 5 \text{ m}$   Allowable Stress: $\sigma_t = 100 \text{ MPa}$ | Solve required core root diameter $d_c$ and select bolt size            |

## 1. Tilting Arm Distance ( $L$ )

Distance measured from base fulcrum edge  $A - A$  of radius  $R = 1 \text{ m}$

$$\begin{aligned} L &= e - R \\ &= 5 - 1 \\ &= 4 \text{ m} \end{aligned}$$

---

## 2. Maximum Tensile Load on Critical Bolt ( $W_t$ )

Circular foundation bolt layout equation for tilting about base edge

$$\begin{aligned} W_t &= \frac{2 \cdot W \cdot L \cdot (R + r)}{n(2R^2 + r^2)} \\ &= \frac{2 \times (100000) \times 4 \times (1 + 0.8)}{8 \times (2(1)^2 + 0.8^2)} \\ &= \frac{800000 \times 1.8}{8 \times 2.64} \\ &= \frac{1440000}{21.12} \\ &= 68.18 \times 10^3 \text{ N} \\ &= 68180 \text{ N} \end{aligned}$$

---

### 3. Required Core Root Diameter ( $d_c$ )

Solve root diameter for allowable stress

$$\begin{aligned}W_t &= \frac{\pi}{4}(d_c)^2 \cdot \sigma_t \\68180 &= \frac{\pi}{4}(d_c)^2 \times 100 \\&= 78.54(d_c)^2 \\(d_c)^2 &= \frac{68180}{78.54} \\&= 868.1 \\d_c &= \sqrt{868.1} \\&= 29.46 \text{ mm}\end{aligned}$$

---

### 4. Standard Thread Selection (Table 11.1)

Select standard coarse thread with  $d_c \geq 29.46 \text{ mm}$

$$\begin{aligned}d_{c, \text{std}} &\geq d_c \\d_c &= 31.093 \text{ mm} \\d &= 36 \text{ mm}\end{aligned}$$

---

### 5. Design Output

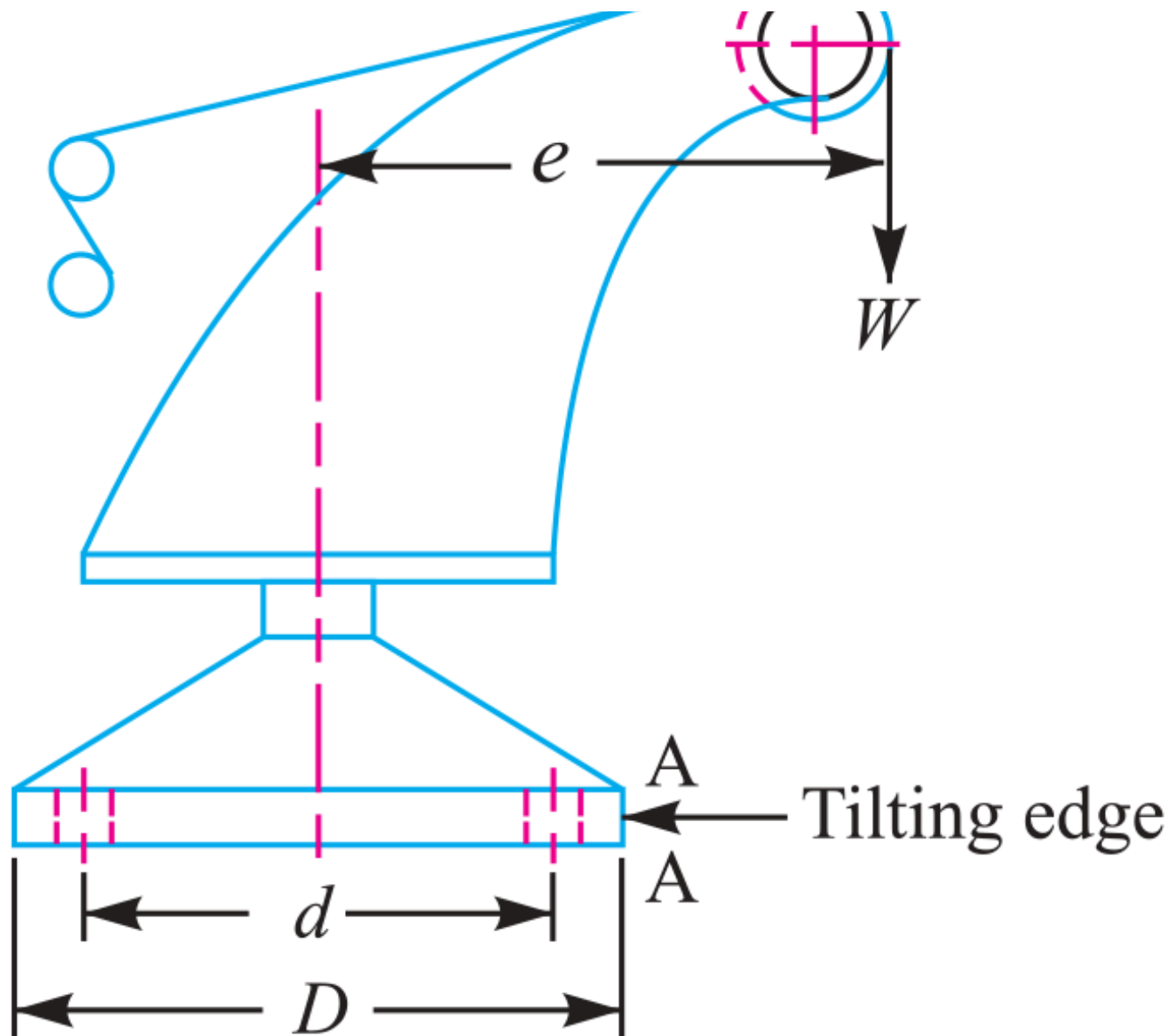
**Select Fastener Designation: M 36 Bolt (d = 36 mm, d\_c = 31.093 mm)**



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## **SECTION 17: Pillar Crane Foundation Flange — MECHANICAL DIAGRAM**

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**Fig 11.42**



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# SECTION 18: Circular Flanged Bearing Base Fastener Sizing

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| System Parameters                                                                                                           | Design Protocol                                               |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Bolt Count: $n = 4$                                                                                                         | For $n = 4$ bolts on circular flange, critical bolt lies at   |
| Bolt Circle Diameter: $d = 500 \text{ mm} \Rightarrow r = 250 \text{ mm}$                                                   | angle $\alpha = \frac{180^{\text{deg}}}{n} = 45^{\text{deg}}$ |
| Bearing Flange Diameter: $D = 650 \text{ mm} \Rightarrow R = 325 \text{ mm}$                                                | Apply circular PCD overturning equation for $W_t$             |
| Overturning Load: $W = 400 \text{ kN} = 400000 \text{ N}$ at arm $L = 250 \text{ mm}$   Stress: $\sigma_t = 60 \text{ MPa}$ | Solve required core root diameter $d_c$                       |

---

### 1. Maximum Tensile Load on Critical Bolt ( $W_t$ )

Overturning equation for  $n = 4$  bolts at  $\alpha = 45^{\text{deg}}$

$$\begin{aligned} W_t &= \frac{2 \cdot W \cdot L \cdot \left[ R + r \cos\left(\frac{180^{\text{deg}}}{n}\right) \right]}{n(2R^2 + r^2)} \\ &= \frac{2 \times (400000) \times 250 \times [325 + 250 \cos 45^{\text{deg}}]}{4 \times (2(325)^2 + (250)^2)} \\ &= \frac{200 \times 10^6 \times [325 + 176.78]}{4 \times 273750} \\ &= \frac{200 \times 10^6 \times 501.78}{1095000} \\ &= 91643 \text{ N} \end{aligned}$$

---

---

## 2. Required Core Root Diameter ( $d_c$ )

Solve root diameter for allowable stress

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$91643 = \frac{\pi}{4}(d_c)^2 \times 60$$

$$= 47.13(d_c)^2$$

$$(d_c)^2 = \frac{91643}{47.13}$$

$$= 1944.5$$

$$d_c = \sqrt{1944.5}$$

$$= 44.1 \text{ mm}$$

---

## 3. Standard Thread Selection (Table 11.1)

Select standard coarse thread with  $d_c \geq 44.1 \text{ mm}$

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 45.795 \text{ mm}$$

$$d = 52 \text{ mm}$$

---

## 4. Design Output

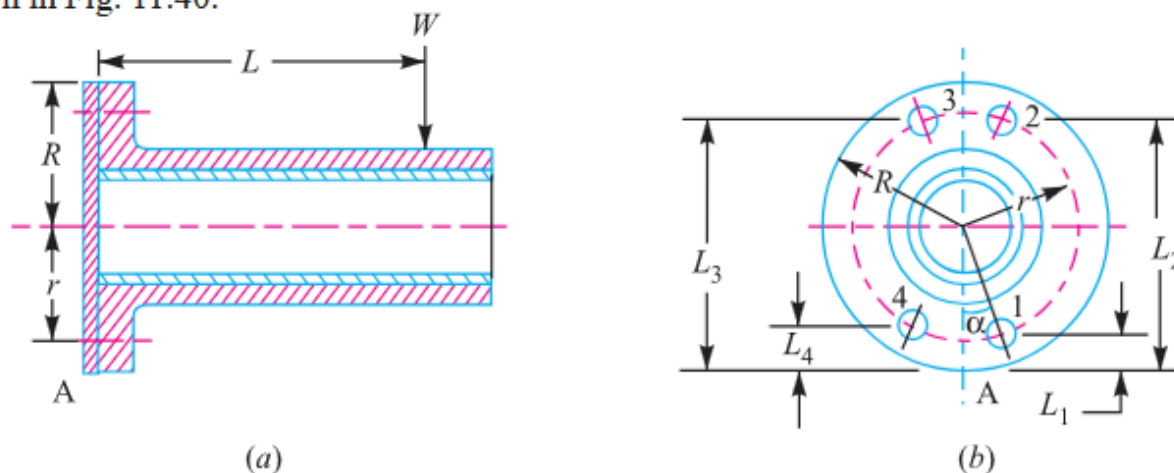
**Select Fastener Designation: M 52 Bolt (d = 52 mm, d\_c = 45.795 mm)**



## SECTION 18: Flanged Bearing Base — MECHANICAL DIAGRAM

### 11.21 Eccentric Load on a Bracket with Circular Base

Sometimes the base of a bracket is made circular as in case of a flanged bearing of a heavy machine tool and pillar crane etc. Consider a round flange bearing of a machine tool having four bolts as shown in Fig. 11.40.



**Fig. 11.40.** Eccentric load on a bracket with circular base.

**Figure 11.40: Circular Flanged Bearing Base PCD Layout**

# SECTION 19: Pillar Crane Circular Foundation 2-Line Overturning Analysis

| System Parameters                                                                  | Design Protocol                                                                 |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Base Diameter: $D = 600 \text{ mm} \Rightarrow R = 300 \text{ mm} = 0.3 \text{ m}$ | Part 1: Analyze line X-X ( $45^{\text{deg}}$ to bolts) to find max distance $e$ |
| PCD: $d = 500 \text{ mm} \Rightarrow r = 250 \text{ mm} = 0.25 \text{ m}$          | Part 2: Analyze line Y-Y (in line with bolts) at same $e$                       |
| Fasteners: $n = 4 \text{ M } 30 \text{ bolts } (A_c = 561\text{mm}^2)$             | to find max induced stress $\sigma_{\text{max}}$                                |
| Overturning Load: $W = 60 \text{ kN} = 60000 \text{ N}$   Allowable                |                                                                                 |
| Stress: $\sigma_t = 60 \text{ MPa}$                                                |                                                                                 |

## 1. Line X-X Analysis ( $45^{\text{deg}}$ Tilting)

Tensile capacity  $P_{\text{cap}} = 561 \times 60 = 33.66 \text{ kN}$ ,  
direct load  $W_{\text{t1}} = \frac{60}{4} = 15 \text{ kN}$

$$\begin{aligned} P_{\text{max}} &= P_{\text{cap}} + W_{\text{t1}} \\ &= 33.66 + 15 \\ &= 48.66 \text{ kN} \end{aligned}$$

$$\begin{aligned} L_1 &= R - r \cos 45^{\text{deg}} \\ &= 0.3 - 0.25(0.7071) \\ &= 0.123 \text{ m} \end{aligned}$$

---

$$\begin{aligned} L_2 &= R + r \cos 45^{\text{deg}} \\ &= 0.477 \text{ m} \end{aligned}$$

$$\begin{aligned} w &= \frac{P_{\text{max}}}{L_2} \\ &= \frac{48.66}{0.477} \\ &= 102 \text{ kN/m} \end{aligned}$$

$$\begin{aligned} M_{\text{res}} &= 2 \cdot w \cdot [L_1^2 + L_2^2] \\ &= 2 \times 102 \times [(0.123)^2 + (0.477)^2] \\ &= 49.4 \text{ kN} \cdot \text{m} \end{aligned}$$

---

## 2. Maximum Load Distance ( $e$ )

Overturning moment  $M_{\text{overturn}} = W(e - R) = 60(e - 0.3)$

$$M_{\text{overturn}} = W(e - R)$$

$$60(e - 0.3) = 49.4$$

$$e - 0.3 = \frac{49.4}{60}$$

$$= 0.8233 \text{ m}$$

$$e = 0.8233 + 0.3$$

$$= \mathbf{1.123 \text{ m}}$$

---

## 3. Line Y-Y Analysis (In-Line with Bolts)

Distances:  $L_1 = 0.05 \text{ m}$ ,  $L_2 = 0.3 \text{ m}$ ,  $L_3 = 0.55 \text{ m}$

$$M_{\text{res}} = w \cdot [L_1^2 + 2L_2^2 + L_3^2]$$

$$= w \cdot [(0.05)^2 + 2(0.3)^2 + (0.55)^2]$$

$$= 0.485 \cdot w$$

$$0.485 \cdot w = 49.4$$

$$w = \frac{49.4}{0.485}$$

$$= 102 \text{ kN/m}$$



---

#### 4. Line Y-Y Maximum Induced Stress ( $\sigma_{\max}$ )

Secondary load  $W_{t2} = w \cdot L_3 = 102 \times 0.55 = 56.1 \text{ kN}$

$$\begin{aligned} P_{\text{net}} &= W_{t2} - W_{t1} \\ &= 56.1 - 15 \\ &= 41.1 \text{ kN} \\ &= 41100 \text{ N} \end{aligned}$$

$$\begin{aligned} \sigma_{\max} &= \frac{P_{\text{net}}}{A_c} \\ &= \frac{41100}{516} \\ &= 79.65 \text{ MPa} \end{aligned}$$

---

#### 5. Design Output

**Final Answers: (1) Maximum Distance  $e = 1.123 \text{ m}$ , (2) Maximum Induced Stress  $\sigma_{\max} = 79.65 \text{ MPa}$**

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## SECTION 19: Pillar Crane 2-Line Analysis — MECHANICAL DIAGRAM

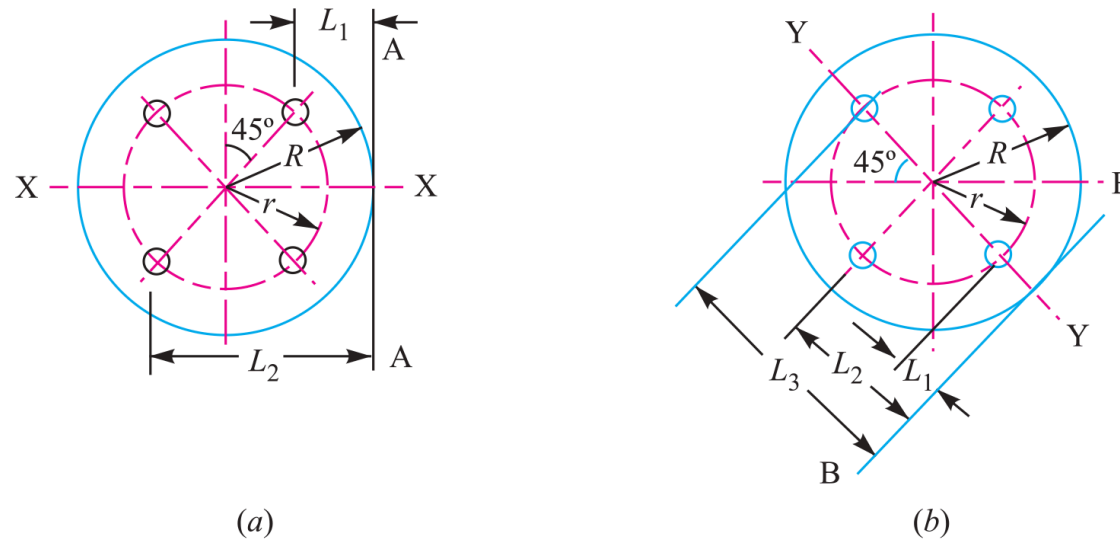


Fig. 11.43

Figure 11.43: 2-Line Overturning Axis Analysis (X-X vs Y-Y)

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# SECTION 20: Solid Forged Bracket under Combined Bending, Torsion & Coplanar Shear

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| System Parameters                                                              | Design Protocol                                                           |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Vertical Load: $W = 13.5 \text{ kN} = 13500 \text{ N}$                         | Part 1: Size arm dia $D$ for combined bending ( $M$ ) and torsion ( $T$ ) |
| Eccentricity: $e = 250 \text{ mm}$                                             | Part 2: Size arm dia $d$ for pure bending ( $M$ )                         |
| Stresses: Tensile $\sigma_t = 110 \text{ MPa}$ , Shear $\tau = 65 \text{ MPa}$ | Part 3: Calculate top bolt tensile load $W_t$                             |
| Flange: 4 bolts in square layout                                               | Part 4: Calculate maximum resultant shear force $W_s$ on bolts            |

## 1. Diameter $D$ for Arm (Combined Bending + Torsion)

$$M = 13500 \times (300 - 25) = 3.7125 \times 10^6 \text{ N} \cdot \text{mm}$$

$$T = 13500 \times 250 = 3.375 \times 10^6 \text{ N} \cdot \text{mm}$$

$$\begin{aligned} T_e &= \sqrt{M^2 + T^2} \\ &= \sqrt{(3.7125 \times 10^6)^2 + (3.375 \times 10^6)^2} \\ &= 5.017 \times 10^6 \text{ N} \cdot \text{mm} \end{aligned}$$

---

$$\begin{aligned}T_e &= \frac{\pi}{16} D^3 \cdot \tau \\5.017 \times 10^6 &= \frac{\pi}{16} D^3 \times 65 \\&= 12.76 D^3 \\D^3 &= \frac{5.017 \times 10^6}{12.76} \\&= 393.2 \times 10^3 \\D &= \sqrt[3]{393.2 \times 10^3} \\&= 73.24 \text{ mm} \\&= \mathbf{75 \text{ mm}}\end{aligned}$$

---

---

## 2. Diameter $d$ for Arm (Pure Bending)

Bending moment  $M = 13500 \times (250 - 37.5) = 2.8688 \times 10^6 \text{ N} \cdot \text{mm}$

$$\begin{aligned}\sigma_t &= \frac{M}{Z} \\ 110 &= \frac{2.8688 \times 10^6}{\frac{\pi}{32}d^3} \\ &= \frac{2.8688 \times 10^6}{0.0982d^3} \\ d^3 &= \frac{29.214 \times 10^6}{110} \\ &= 265.58 \times 10^3 \\ d &= \sqrt[3]{265.58 \times 10^3} \\ &= 64.28 \text{ mm} \\ &= \mathbf{65 \text{ mm}}\end{aligned}$$

---

## 3. Tensile Load on Each Top Bolt ( $W_t$ )

Tilting distances  $L_1 = 37.5 \text{ mm}$ ,  $L_2 = 237.5 \text{ mm}$

$$\begin{aligned}M_{\text{res}} &= 2 \cdot w \cdot (L_1^2 + L_2^2) \\ &= 2 \cdot w \cdot (37.5^2 + 237.5^2) \\ &= 115625 \cdot w\end{aligned}$$

$$\begin{aligned}
 M_{\text{overturn}} &= W \cdot L \\
 &= 13500 \times 300 \\
 &= 4.05 \times 10^6 \text{ N} \cdot \text{mm} \\
 w &= \frac{4.05 \times 10^6}{115625} \\
 &= 35.03 \text{ N/mm}
 \end{aligned}$$

$$\begin{aligned}
 W_t &= w \cdot L_2 \\
 &= 35.03 \times 237.5 \\
 &= 8320 \text{ N} \\
 &= \mathbf{8.32 \text{ kN}}
 \end{aligned}$$

#### 4. Maximum Shearing Force on Each Bolt ( $W_s$ )

Primary shear  $W_{s1} = \frac{13500}{4} = 3375 \text{ N}$ , radial dist  
 $r_i = \sqrt{100^2 + 100^2} = 141.4 \text{ mm}$

$$\begin{aligned}
 W_{s2} &= \frac{W \cdot e \cdot r_i}{\sum r_i^2} \\
 &= \frac{13500 \times 250 \times 141.4}{4 \times (141.4)^2} \\
 &= 5967 \text{ N}
 \end{aligned}$$

$$W_s = \sqrt{W_{s1}^2 + W_{s2}^2 - 2W_{s1}W_{s2}\cos\theta}$$

$$W_s \quad (\text{Bolts 1 \& 4}) = 4303 \text{ N}$$

$$W_s = \sqrt{W_{s1}^2 + W_{s2}^2 + 2W_{s1}W_{s2}\cos\theta}$$

$$W_s \quad (\text{Bolts 2 \& 3}) = 8687 \text{ N}$$

## 5. Design Output

**Final Answers: (1) D = 75 mm, (2) d = 65 mm, (3) Top Bolt W<sub>t</sub> = 8.32 kN, (4) Max Shear Force = 8687 N**

## SECTION 20: Solid Forged Bracket — MECHANICAL DIAGRAM

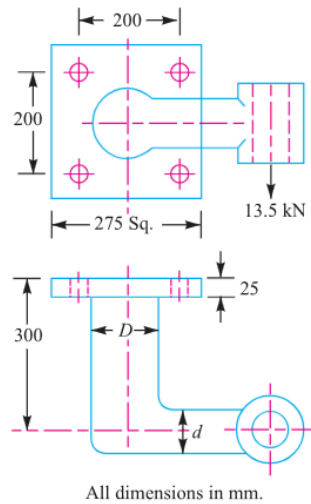


Fig. 11.45

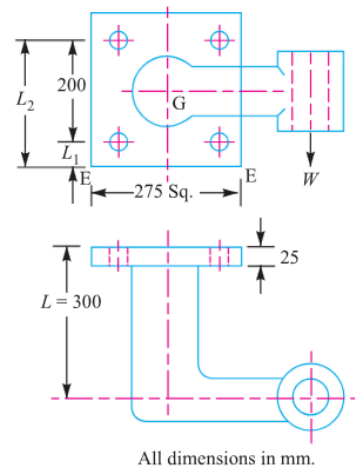


Fig. 11.46

Figure 11.45 & 11.46: Solid Forged Bracket Structure

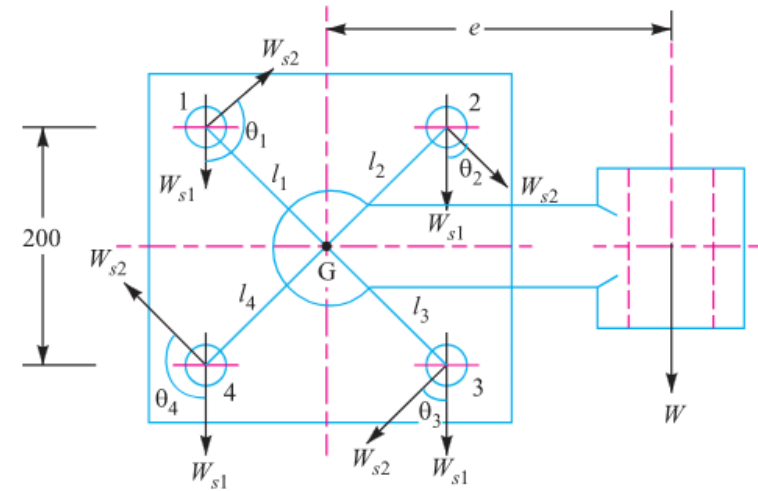


Fig. 11.47

Figure 11.47: Vector Shear Load Diagram on Fastener Group